

Intermingling Science and Religion in Writings from the Scientific Revolution

Typically, the sixteenth and seventeenth centuries are remembered as the moment when natural philosophers turned away from scripture and toward modern mechanisms of scientific thinking. It would appear that mythology and religion, under this cultural context, should be rationalized out of existence, or at least out of academic discourse. Yet in studying texts from the history of scientific writing during this period, I find that natural empiricism and divine explanations often intermingled, giving us a window into the many ways of knowing that coexisted at the time. To explore this, I look at letters and treatise written by three figures during this period: Galileo, Robert Boyle, and Ulisse Aldrovandi, to show how scientific and religious thought intermingled in the history of scientific writing. Finally, I compare them to Hippocrates *Sacred Disease* and to a modern example to situate how these ways of thinking compare to different points of time in the history of scientific writing. Altogether, these texts not only complicate our narrative of linear progress away from religion and toward pure empiricism, but make legible the possibilities for a coexistence of multiple different ways of knowing.

Slide 1: Galileo

First, I look at the classic case of Galileo. He is remembered as the heretic willing to challenge the dogmatic church with his theory of Heliocentrism, which led to his condemnation from the Catholic Church; However, this narrative of faith vs science is oversimplified, as Galileo was also a devout Catholic (Zanatta et al), who tied religious thinking into his scientific explanations of the universe. This is demonstrated in his 1613 letter to former pupil Benedetto Castelli, a mathematics professor and Benedictine monk. This letter, importantly, is also credited as launching the Church's inquisition, after it was circulated widely so other astronomers could

learn from it (Camerota et al). In this letter, Galileo made a complex theological argument that tied his scientific theories together with theories of the divine, instead of separating them. You will see the original letter in Italian, on my slides¹, but thanks to an English translation in Finocchiaro's historical account of Galileo we can see the specifics of his argument (Galilei, "Letter to Benedetto"); Galileo writes, "the Holy Scripture and nature both equally derive from the divine Word, the former as the dictation of the Holy Spirit, the latter as the most obedient executrix of God's commands" (Finocchiaro and Galilei 50), showing that Galileo understood scripture and nature as two expressions of the same divine truth. He believed his theory of heliocentrism was not heretical or against the bible, and interpreted his findings in the light of complex theological arguments about scripture: he argues that "limit[ing] oneself always to the literal meaning of the words" is not a wise theological position, and this his opponents using it to debunk heliocentrism is unwise, as "there would thus emerge not only various contradictions but also serious heresies and blasphemies, and it would be necessary to attribute to God feet, hands and eyes..." demonstrating how he situated his scientific theories within the debate between strict or non-strict interpretationism of the bible, instead of arguing it as a wholly unrelated scientific claim which stood on the outside of this religious discourse (49-50).

His theological arguments deepen in complexity as one reads on. After arguing against literal interpretation, he shows that his claims could still hold up if one "concede[s]" that scripture "should be taken precisely in their literal meaning," by analyzing a passage from Joshua where God stops the sun and it results in the lengthening of the day. He compares this passage to the implications of the heliocentric vs geocentric model, and explains that the result of lengthening the day is more consistent with his model, making a direct argument that there is support for his theory in the actual text of the bible (52-54). Moreover, he addresses the letter to

¹ See appendix for select images that would be on the slides

the “Very Reverend Father and My Most Respectable Sir” Castelli, and signs off the letter, “pray[ing] the Lord to give you happy holidays,” implying a respect and devoutness that the specific epistolary format of this source helps us see (49, 57). The implication is that even for Galileo, who faced intense scrutiny from some members of the church for his theories, a separation between religious thinking and scientific thinking wasn’t wholly incompatible, and in fact, he believed religion reinforced his heliocentric theory.

Galileo’s religious invocations do not negate the intense scrutiny he received from certain members of the clergy in denouncing his work— in fact, one Dominican friar would use the same passage from Joshua in an attempt to disprove Galileo, just the next year (Zanatta et al). However, this same biblical passage is mobilized to make a different argument, which demonstrates that it is interpretation that is the difference between seeing faith as incompatible with religion, and as able to coexist with it in explaining the universe. In this way, Galileo’s letters complicate the simple faith-versus-science story, revealing a thinker for whom theological reasoning and empirical observation were mutually reinforcing.

Slide 2: Boyle

Moving on from Galileo, I encourage you to turn to the intertwined empirical and theological arguments of Robert Boyle, exalted as a founder of modern chemistry and of Boyle’s Law fame (Williams). Despite his modern methods, Philosopher of Science Thomas Sukopp draws on treatise and correspondence from Boyle to map how his religion influenced his science, and argues that he “aspir[ed] to harmonise the religious and supernatural features of the world with scientific findings” (Sukopp 62). Building from Sukopp’s analysis, I demonstrate how Boyle’s religious and scientific intertwinement functioned similarly to Galileo’s own claims. In

making this comparison, I aim to establish this intermingling of these modes of thinking went beyond a single person in the scientific revolution. For example, in a letter to Henry Stubbe, on 9 March 1666,² Boyle states: "I am far from believing that any Mechanicall of Phisicall hypothesis will make out those supernaturall Phaenomena without having recourse to the Miraculous Interposition of God" (Sukopp 63, qtd Boyle). In other words, he argues that scientific theories likely cannot "make out" or be used to understand God, but in saying so, he is not denying God's existence, just that the proof of faith is outside the bounds of physical hypothesis. This statement is similar to Galileo's claim of two truths, which do not run parallel to each other and cannot be reconciled in a literal fashion, but which both function to describe reality accurately. Moreover, in *The grounds for an Excellence of the Corpuscular or Mechanical Philosophy* (1674), Boyle argues for a universe and natural laws that were set in motion by God: he writes that his theory

distinguishes (a) the ultimate origins of things from (b) the subsequent course of nature, and teaches concerning (a) that God not only gave motion to matter but also at the outset guided the various motions of its parts in such a way that he contrived them into the world he planned for them to compose—a world supplied with seminal principles [see Glossary] and structures or models of living creatures—and established the rules of motion and the order amongst bodies that we ordinarily call 'the laws of nature'... (Sukopp 65, qtd Boyle)

God, and with the laws of motion being settled and then upheld by his unceasing concurrence, the natural laws he was discovering operated within a framework ultimately sustained by divine will, but does not require active intervention (Sukopp 64, qtd Boyle). This argument is similar to Galileo's argument that nature is the "executrix" of God's will, demonstrating another throughline in how religious and empirical philosophy appeared in Scientific Revolution thinkers.

² Note on a letter to Henry Stubbe, on 9 March 1666: Though we discussed this in class, I was unable to request the Boyle correspondence from Wilson Library in time to get this final project complete, so I used a proxy through a secondary source. In the future, if I continued this project, I would absolutely request this information so I could see the actual source and analyze it!

However, the Scientific Revolution was not monolithic, and these thinkers did not always agree. In my own analysis of Boyle's *Corpuscular Treatise*, I find that despite ultimately arguing for a natural laws that God put into place, Boyle acknowledges a continuum of explanations, both divine and completely natural, that speaks to the ability of religion and the organized skepticism of science to coexist. Returning to his discussion of motion and matter, he writes, "Either both of them were immediately created by God or, if matter is eternal and thus was never created (as some hold to be the case) it must be the case that -motion was produced by some immaterial supernatural agent, or -motion is something that the moving matter just naturally produces itself" (Boyle, "Corpuscular or Mechanical Philosophy" 3). The latter options, that matter "was never created" or that motion "just naturally produces itself" provides an alternative to his other options which invoke the role of the divine in setting all laws in motion. In considering both religious and mechanical explanations as open questions to be further considered, despite a broader aspiration towards reconciling religious and scientific doctrines, in and of itself blends scientific empirical tactics, such as skepticism, with a broader openness to religion. A willingness to engage with either belief system and hold both as options simultaneously is a different, but still vital way in which these two belief systems can be seen as coexisting.

Slide 3: Aldrovandi

Finally, I consider Ulisse Aldrovandi's 1642 *Monstrorum Historium* or, in English, *History of Monsters* (Aldrovani). As the provocative name implies, this work is not lacking in references to the monstrous, mythological and religious, but it also presents salient information on medical conditions and birth defects, detailed descriptions of the anatomy of the uterus, and

both descriptions and medical advice for childbirth complications. Aldrovandi, similarly to Galileo and Boyle, positioned medicine and biology within a larger context of divinity, proving to be another example of how these ways of knowing coexisted in the Scientific Revolution.

For example, consider Aldrovandi's depiction of the centaur— a figure from Greek/Roman mythology which was half man (the upper half) and half horse (the lower half), and other human animal hybrids like satyrs (upper half man, lower half goat). Deviating from Boyle and Galileo, Aldrovandi relies on a number of Greek and Roman writers and their accounts to describe the centaur, however, this is still a religious invocation and thus illuminates a new facet of religious consideration that played a role in Scientific Revolution texts. Indeed, this was in line with how most of these histories were created: according to Historian of Science Anne Bitbol-Hespériès, most physicians did not see the creatures they wrote about directly (which they attributed to their rarity and the fact that these monsters didn't live long) so instead they drew their information from the classics, biblical writings, and other ancient writers such as the Greeks and Romans (Bitbol-Hespériès 47).

Despite his reliance on mythology to understand many of these creatures, in zooming into the case of the centaur and satyr, we see how Aldrovandi ultimately puts mythological conclusions about human-animal hybrids in discourse with biological empiricism. This results in him rejecting many Greek claims, which would seem to be in line with a move towards scientific rationalist methods, yet his conclusion situates these creatures within Christian thought, showing a continued reliance on utilizing religious explanatory systems. Fascinatingly, he mobilizes biological empiricism to trade one religious explanatory system for another. In the case of the centaur, he first reviews the propositions by other writers that these creatures came from sexual intercourse between horses and humans (ie Pliny and Plutarch). He is unconvinced, citing other

medical writers and thinkers who worked on interspecies reproduction, namely, the observation that body sizes and gestation lengths must be somewhat comparable for reproduction to occur (Aldrovandi [translated] 9). He uses these biological writings to conclude that most sightings of centaurs were simply visual illusions from riders and their horses galloping quickly past confused onlookers, but he does believe there have been veritable sightings of centaurs, such as the case of St. Anthony's sighting one, in which he concludes that a demon took the form of Centaur to trick him (42). He comes to a similar conclusion for the Satyrs', who he concludes were often just the devil in disguise, trying to lead humanity astray, which is demonstrated by the fact that these "malignant spirits" vanished after the death of Christ redeemed humanity (34). Though he trades one religion for another, Aldrovani's discourse brings in the role of the divine as an overarching theology with biological details grounded in empiricism.

While some religious thinking in science— like Aldrovandi's conclusions about the devil in science has ultimately been replaced, it is hard to argue against Boyle's law and Galileo's heliocentrism, showing how religion and science has historically been able to coexist in furthering our knowledge of the natural world.

Slide 4: Against a narrative of progress

Having drawn out the ways in which science and religion intermingled during this crucial period, I want to be careful here, because comparatively, we seem to invoke less religion now in our science. A natural conclusion then, would be that religion slowly faded out until it was less and less a part of science. Our narrative of scientific progress would tell you that in the past, there must have been even more religion in science too. To disprove this narrative of linear progress, I want to bring up an example from two thousand years before the Scientific

Revolution: Hippocrates' *On the Sacred Disease*, which referred to what we call epilepsy. There, Hippocrates argued against considering this disease sacred, writing, one "cannot consider [diseases like epilepsy] either sacred or divine; for when they are removed by such purifications and by such treatment as this, there is nothing to prevent the production of attacks in men by devices that are similar. If so, something human is to blame, and not godhead" (Hippocrates "Sacred Disease" [trans] 145). In other words, he states that diseases which are caused by natural objects can be cured by natural objects, and diseases which are caused by the divine must be cured by the divine. Since epilepsy can be by "purifications and treatment[s]" that are not divine, "human" is to blame. This quote does not deny the broader context of religion and the existence of the Gods, which is reminiscent of the Enlightenment writers, despite the time gap. On the other hand, consider an example from today: original antigenic sin, a term coined in 1960 to describe how the body's immune system remembers its earliest infections, like the flu, was described in terms of a Christian Drama (Apple). Moreover, the title, the *Doctrine of OAS*, invokes rules and mechanisms through the word "doctrine" which is almost always associated with religion, demonstrating how religion continues to influence our conceptualization of the sciences (Francis). Looking at these examples, we see that there is a great deal of intermingling between scientific thought and religious thought at many discrete points in time, and not simply in one direction.

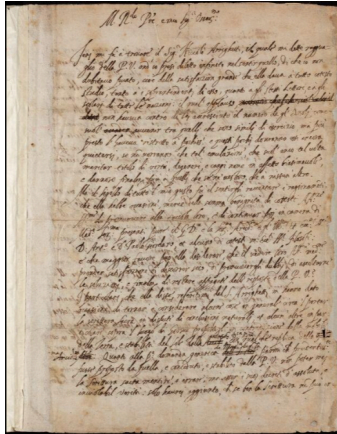
Slide 5: Conclusion: Religion vs Science?

I hope that by looking at the history of writing in the scientific revolution, and briefly to other points in time, I have demonstrated the ways in which religious and scientific thought mingle in producing our understanding of the world. Whether directly, as when Aldrovandi attributes real centaur sightings to Satan in disguise, or more subtly, as when Boyle's and

Galileo's natural laws operate within a universe sustained by God, these invocations have shaped the way in which the natural world is understood, and cannot be ignored. Especially in a time when science and religion are often pitted against each other in health policy debates and in making important decisions about how we live our lives, I believe it is important to acknowledge how different ways of knowing can coexist. The history of writing during the Scientific Revolution is a tool that makes this coexistence legible to us, and will hopefully help us build our way to more tolerance and understanding for how diverse ways of knowing can intermingle.

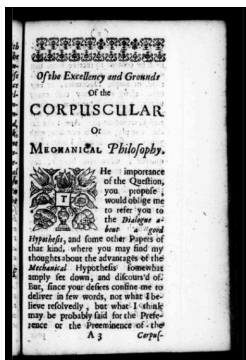
Appendix:

SLIDE 1 IMAGES:



Galileo Galilei, “Letter from G[alileo] G[alilei] to Padre Benedetto Castelli,” dated at Florence, 21 December 1613

SLIDE 2 IMAGES:



Robert Boyle, The Grounds for and Excellence of the Corpuscular or Mechanical Philosophy, 1674. Part of a larger work on The Excellency of a variety of topics, 1674, on Internet Archive.

SLIDE 3 IMAGES:

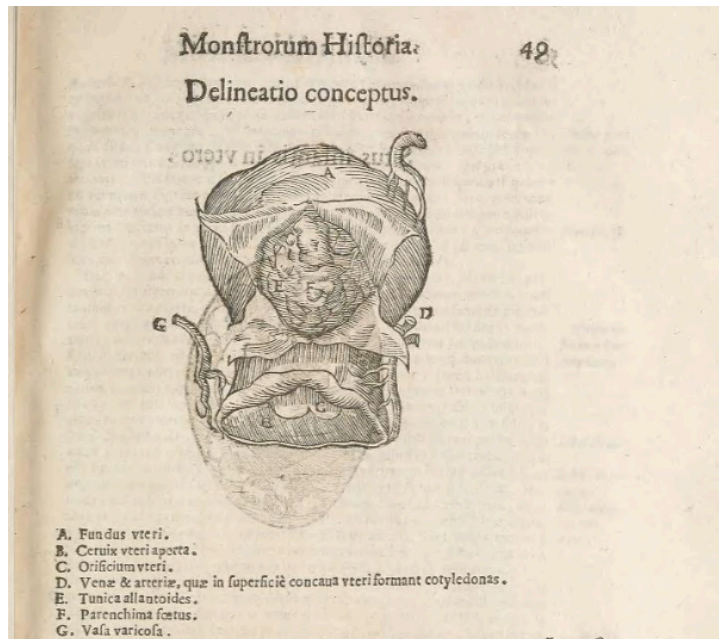


Illustration of a uterus from Ulisse Aldrovandi's *Monstrorum Historium*- note that the key denotes parts with the same Latin terminology we use today, such as Fundus, Cervix etc- 1642, Image from Internet Archive.



Illustration of a centaur from Ulisse Aldrovandi's *Monstrorum Historia*, 1642, Image from Internet Archive.

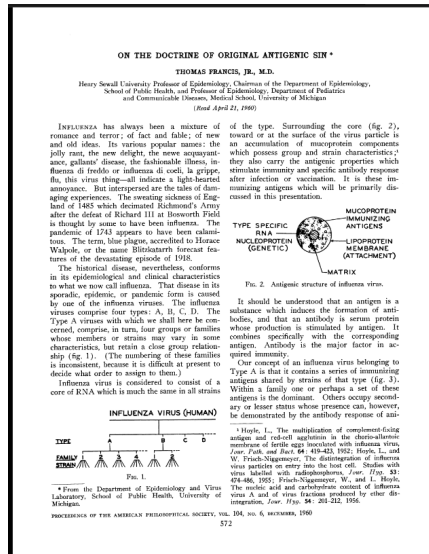


Illustration of two different types of satyrs from Ulisse Aldrovandi's *Monstrorum Historia*, 1642, Image from Internet Archive.

SLIDE 5 IMAGES:



Image 8 of Greek Manuscripts 1387. Hippocrates. From Library of Congress



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